

## 1. Secure Data Transmission using Block Cipher

### A. Explain why this issue occurs.

In Electronic Codebook (ECB) mode, each plaintext block is encrypted independently using the same encryption key. As a result, if two plaintext blocks are identical, their corresponding ciphertext blocks will also be identical. When repeated financial data is transmitted, attackers can observe these repeating ciphertext patterns and gain information about the structure of the original data. Although the actual content remains encrypted, these visible patterns reduce confidentiality and make ECB unsuitable for sensitive financial communications.

### B. Suggest a more secure block cipher mode and justify your choice.

Cipher Block Chaining (CBC) mode is a more secure choice.

In CBC mode, each plaintext block is XORed with the previous ciphertext block before encryption. The first block uses an Initialization Vector (IV). This means that identical plaintext blocks produce different ciphertext blocks because each block depends on the previous encrypted block.

#### Justification:

- Eliminates repeated ciphertext patterns.
- Provides better confidentiality for repeated financial data.
- Makes statistical and pattern analysis attacks much more difficult.
- Widely used for secure data transmission.

**Decision:** CBC mode is recommended because it effectively hides patterns in repeated financial data.

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## 2. Choosing an Encryption Algorithm for Banking System

### A. Identify the most suitable algorithm.

Among DES, 3-DES, and Blowfish, the most suitable algorithm is Blowfish.

### B. Justify your choice based on security strength and efficiency.

#### DES

- Uses a 56-bit key.
- Considered insecure because modern computers can break it through brute-force attacks.
- Not suitable for banking applications.

#### 3-DES

- Uses a 168-bit key.
- More secure than DES.
- However, it encrypts data three times, making it slower and computationally expensive.

## Blowfish

- Supports key sizes up to 448 bits.
- Provides strong security.
- Faster than both DES and 3-DES.
- Efficient for software implementation.

**Decision:** Blowfish is recommended because it provides strong encryption while maintaining good performance, making it suitable for secure banking transactions.

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## 3. Key Exchange in an Unsecured Network

### A. Explain how Diffie-Hellman key exchange works.

Diffie-Hellman allows two users to establish a shared secret key over a public network without transmitting the secret key directly.

#### Steps:

1. Both users agree on public values **p** (a prime number) and **g** (a generator).
2. User A selects a private key and computes a public key.
3. User B selects a different private key and computes another public key.
4. Both users exchange their public keys.
5. Each user combines the received public key with their own private key to compute the same shared secret key.
6. The shared secret is then used for secure communication.

### B. Identify a security threat and suggest a solution.

**Threat:** Man-in-the-Middle (MITM) Attack.

An attacker intercepts the exchanged public keys and replaces them with their own. As a result, both users unknowingly establish separate secret keys with the attacker instead of each other.

#### Solution:

- Use Digital Signatures to authenticate public keys.
- Use Digital Certificates issued by a trusted Certificate Authority (CA).
- Implement authenticated protocols such as TLS/SSL.

**Decision:** Authenticated Diffie-Hellman using digital certificates effectively prevents Man-in-the-Middle attacks.

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## 4. Public Key Cryptography for Email Security

### A. Describe how RSA is used for encryption and decryption.

RSA uses a pair of keys:

- **Public Key** – shared with everyone.
- **Private Key** – kept secret by the owner.

### **Encryption Process**

1. The sender encrypts the email using the recipient's public key.
2. The encrypted email is transmitted over the network.
3. Only the recipient can decrypt it using their private key.

### **Decryption Process**

1. The recipient receives the encrypted email.
2. The private key is used to decrypt the ciphertext.
3. The original message is recovered securely.

This ensures that only the intended recipient can read the email.

### **B. Explain the role of key distribution and management.**

Effective key management is essential for RSA security.

It includes:

- Secure generation of public and private keys.
- Safe storage of private keys.
- Secure distribution of public keys using digital certificates.
- Regular key updates and revocation if keys are compromised.

Proper key management prevents unauthorized access and protects email confidentiality.

**Decision:** RSA combined with secure key management provides strong protection for email communication.

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## **5. Cryptography for IoT Devices**

### **A. Analyze the constraints of IoT devices.**

IoT devices typically have:

- Limited processing power.
- Low memory capacity.
- Limited battery life.
- Restricted storage.
- Low communication bandwidth.

Because of these limitations, cryptographic algorithms must be lightweight and energy-efficient.

### **B. Recommend a suitable cryptographic approach and justify your answer.**

Elliptic Curve Cryptography (ECC) is the most suitable choice.

**Justification:**

- Provides security equivalent to RSA using much smaller key sizes.
- Requires less computation.
- Consumes less memory.
- Reduces power consumption.
- Faster key generation and encryption operations.
- Well suited for battery-powered IoT devices.

**Decision:** ECC is recommended because it offers strong security while minimizing computational cost, memory usage, and power consumption, making it ideal for smart home IoT systems.

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**Conclusion**

| Question              | Recommended Solution                                    | Reason                                                                       |
|-----------------------|---------------------------------------------------------|------------------------------------------------------------------------------|
| 1. Block Cipher Mode  | CBC Mode                                                | Hides repeated plaintext patterns and improves confidentiality.              |
| 2. Banking Encryption | Blowfish                                                | Strong security with better performance than DES and 3-DES.                  |
| 3. Diffie-Hellman     | Authenticated Diffie-Hellman (TLS/Digital Certificates) | Prevents Man-in-the-Middle attacks.                                          |
| 4. Email Security     | RSA with Proper Key Management                          | Ensures secure encryption, decryption, and authentication.                   |
| 5. IoT Security       | Elliptic Curve Cryptography (ECC)                       | Low power, low memory, and strong security for resource-constrained devices. |